

MATERIAL TEST REPORT

HFW STEEL PIPE GENERAL INFORMATION																	
ITEM	O.D.(in)	Wall Thickness (in)	Length (ft)	Specification(s) or Reference and Grade	Coating	End Finish	NDT Weld Seam	NDT Pipe Body	Drift	Visual and Dimension Control	Flattenning Test	Min Collapse Pressure	Hydrostatic Test		Heat Treatment	Workorder	Date
													Pressure (Psi)	Hold Time (sec)			
1	12.750	0.500	38.00-47.00	API 5L 46th ed X52M PSL2	Zero VOC Clear Mill Coating	30 Beveled End	UT: OK N 10: 0.125in through hole	API 5L - E3.2 Pipe End Inspection - Welded Pipe	N/A	OK	OK	N/A	1740	5	Weld Seam Annealed Min 1598° F	2501W018217	2/2/2025

			Chemical Analysis															
ITEM	Heat No	Lot No		C (%)	Mn (%)	P (%)	S (%)	Si (%)	Al (%)	Cu (%)	Ni (%)	Cr (%)	Mo (%)	V (%)	Nb (%)	Ti (%)	B (%)	Ceq
			Max	0.220	1.400	0.025	0.015	0.450		0.200	0.200	0.100	0.150	0.050	0.050	0.040		0.250
			Min											Nb + V + Ti <= 0.15%				
			Ceq (sum)	C (%)	Mn % / 20			Si % / 30		Cu % / 20	Ni % / 60	Cr % / 20	Mo % / 15	V % / 10			5 X B%	
1	NCX-2147069	N/A	Product	0.046	1.100	0.011	0.000	0.200	0.027	0.095	0.042	0.055	0.013	0.001	0.026	0.011	0.0003	0.118
			Skelp	0.050	1.100	0.011	0.000	0.200	0.029	0.095	0.043	0.055	0.013	0.001	0.026	0.011	0.0003	0.122
			Heat	0.040	1.080	0.012	0.001	0.180	0.030	0.090	0.030	0.040	0.010	0.005	0.034	0.011	0.0000	0.108
	NCX-2147070	N/A	Product	0.048	1.090	0.011	0.000	0.200	0.025	0.093	0.041	0.053	0.013	0.001	0.026	0.011	0.0003	0.120
			Skelp	0.048	1.090	0.011	0.000	0.198	0.024	0.092	0.042	0.053	0.013	0.001	0.026	0.011	0.0003	0.119
			Heat	0.040	1.110	0.013	0.001	0.190	0.030	0.100	0.040	0.060	0.010	0.006	0.035	0.012	0.0000	0.112
	NCX-2247497	N/A	Product	0.058	1.100	0.009	0.000	0.191	0.026	0.093	0.034	0.044	0.011	0.000	0.023	0.011	0.0003	0.129
			Skelp	0.063	1.110	0.009	0.000	0.193	0.026	0.094	0.034	0.044	0.012	0.001	0.024	0.011	0.0004	0.135
			Heat	0.050	1.090	0.011	0.001	0.170	0.030	0.100	0.030	0.040	0.010	0.004	0.031	0.012	0.0000	0.119

			Mechanical Properties																		
			Transverse Weld									Transverse Body									
			Seam Weld, 8" overall, 2.250" reduction area, 2" gauge length. Flat specimen		Impact Test at 32° F (ft/lb)					Pipe Body, 10.250" over all length, 2.250" length reduction area, 2" gauge length, T180, Flat specimen			Impact Test at 32° F (ft/lb)/ 3T90								
ITEM	Heat No	Lot No	Tensile Strength (PSI)	Vickers Hardness HV10	Sample Size:	Full				Yield Strength (PSI)	Tensile Strength (PSI)	Elongation (%)	Vickers Hardness HV10	Sample Size:	Full						
						Specimen 1 Impact Energy/ Shear	Specimen 2 Impact Energy/ Shear	Specimen 3 Impact Energy/ Shear	Average Impact Energy						Average Shear (%)	Specimen 1 Impact Energy/ Shear	Specimen 2 Impact Energy/ Shear	Specimen 3 Impact Energy/ Shear	Average Impact Energy	Average Shear (%)	
			Max		240							76,900				240					
			Min	66,700		20	20	20				52,200	66,700			20	20	20			
1	NCX-2147069	N/A	Product	72,224	161	406.3 / 100	406.3 / 90	391.2 / 90	401	93	60,930	69,255	48	161	402.2 / 90	402.6 / 90	402.2 / 90	402	90		
1	NCX-2147070	N/A	Product	76,180	165	398.6 / 100	399.7 / 100	403.9 / 90	401	96	55,379	68,350	45	159	406.3 / 90	406.3 / 90	406.3 / 90	406	90		
1	NCX-2247497		Product	74,557	169	424 / 100	448.8 / 100	435.1 / 100	436	100	65,970	70,643	43	164	451.1 / 90	436.9 / 90	445.9 / 100	445	93		

		Impact Test at -20° F (ft/lb)				
Heat No	Sample Size:	Full				
	Specimen 1 Impact Energy/ Shear	Specimen 2 Impact Energy/ Shear	Specimen 3 Impact Energy/ Shear	Average Impact Energy	Average Shear (%)	
	20	20	20			
	NCX-2147069	406.3 / 100	406.3 / 90	371.2 / 90	395	93
	NCX-2147070	398.6 / 100	399.7 / 100	403.9 / 90	401	97
NCX-2247497	424.0 / 100	448.8 / 100	435.1 / 100	436	100	

	Impact Test at -20° F (ft/lb)/ 3T90					
Heat No	Sample Size:	Full				
	Specimen 1 Impact Energy/ Shear	Specimen 2 Impact Energy/ Shear	Specimen 3 Impact Energy/ Shear	Average Impact Energy	Average Shear (%)	
	20	20	20			
	NCX-2147069	402.2 / 90	402.6 / 90	402.2 / 90	402	93
	NCX-2147070	406.3 / 90	406.3 / 90	406.3 / 90	406	90
NCX-2247497	451.1 / 90	436.9 / 90	445.9 / 100	445	93	

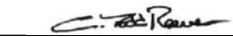
Coil Source: Nucor Steel (7301 East County Road 142, Blytheville, AR 72315)
Melted and Manufactured in USA

Steel Source: Nucor Steel (7301 East County Road 142, Blytheville, AR 72315)
Melted and Manufactured in USA

WE CERTIFY THE ABOVE MATERIAL HAS BEEN MANUFACTURED, PROCESSED, SAMPLED, TESTED, AND INSPECTED TO MEET THE REQUIREMENTS OF THE STANDARD IN REFERENCE.
DETERMINATION OF MECHANICAL TESTING ARE ACCURATE AND TRACEABLE TO NIST AND CONFORM TO APPLICABLE TESTING METHODS.
CONFORMS TO NACE MR-0175 FOR HARDNESS ONLY AS IT APPLIES TO API PRODUCTS. MAX HARDNESS DOES NOT EXCEED RC22.
NOT VALID WITH INADEQUATE USE OF PIPE.
THIS INSPECTION CERTIFICATE IS IN ACCORDANCE WITH API 5L 46TH EDITION.

Borusan Pipe

Todd Reeves /



VP OF QUALITY ASSURANCE